PRIORITIZATION OF THE SUB/MICRO WATERSHEDS BASED ON SOIL LOSS POTENTIALITY USING MORPHOMETRIC ANALYSIS: A CASE STUDY ON KANGSABATI RIVER BASIN IN PURULIA, WEST BENGAL

1. INTRODUCTION:

Soil erosion, a significant environmental issue, results from the detachment and transportation of soil particles by water or wind. This process can lead to soil degradation, reduced agricultural productivity, and sedimentation of water bodies. To effectively manage soil erosion, it is crucial to identify areas with high soil loss potential and prioritize them for conservation measures.

Morphometric analysis, a quantitative technique that involves the measurement and analysis of landform characteristics, provides valuable insights into the hydrological processes and erosion susceptibility of a watershed. By examining parameters like drainage density, stream order, relief ratio, and bifurcation ratio, we can assess the potential for soil erosion in different sub-watersheds.

Drainage density, the total length of stream channels per unit area, is a key indicator of erosion potential. Higher drainage density often indicates a steeper slope and higher erosion risk. Stream order, a hierarchical classification of streams based on their branching pattern, can also provide insights into erosion potential. Higher-order streams tend to have greater erosive power and can transport more sediment.

Relief ratio, the difference in elevation between the highest and lowest points in a watershed divided by the horizontal distance, is another important morphometric parameter. A higher relief ratio indicates a steeper slope and greater erosion potential. Bifurcation ratio, the ratio of the number of streams of a given order to the number of streams of the next higher order, [1] can also be used to assess erosion susceptibility. A higher bifurcation ratio suggests a more dendritic drainage pattern, which can be associated with higher erosion rates.

By integrating morphometric analysis with other relevant factors like soil type, land use, and climate data, it is possible to develop a comprehensive assessment of soil loss potentiality. This information can be used to prioritize sub-watersheds for targeted conservation measures, such as afforestation, terracing, and contour farming.

In this study, we will employ morphometric analysis to assess the soil loss potentiality of sub-watersheds within a specific region. By understanding the spatial variability of erosion risk, we can develop effective strategies to mitigate soil erosion and protect valuable soil resources. Additionally, by incorporating other relevant factors like soil erodibility, rainfall erosivity, and land use practices, we can refine our assessment and identify areas that are most vulnerable to soil erosion. Furthermore, integrating remote sensing and GIS techniques can enhance the accuracy and efficiency of morphometric analysis. By using high-resolution satellite imagery and digital elevation models, we can generate detailed maps of landform characteristics and identify areas with high erosion potential.

2. STUDY AREA:

The Kangsabati River Basin, located in the Purulia district of West Bengal, India, is a geographically diverse region characterized by undulating hills, valleys, and riverine landscapes. The basin lies between the latitudes 23° 09' N to 23° 37' N and longitudes 86° 00' E to 86° 30' E. The Kangsabati River, the principal river of the basin, originates in the Chotanagpur Plateau and flows through the district, eventually joining the Damodar River.

The basin is predominantly covered by lateritic soil, which is highly susceptible to erosion due to its low fertility and poor structure. The region experiences a tropical monsoon climate, characterized by distinct wet and dry seasons. The monsoon season, from June to September, brings heavy rainfall, which can trigger intense erosion, particularly in areas with steep slopes and fragile soils.

The basin is home to a diverse range of land use and land cover types, including agricultural land, forest, and barren land. Agricultural practices, such as shifting cultivation and overgrazing, can exacerbate soil erosion. Deforestation and forest degradation further contribute to the problem by removing the protective vegetation cover.

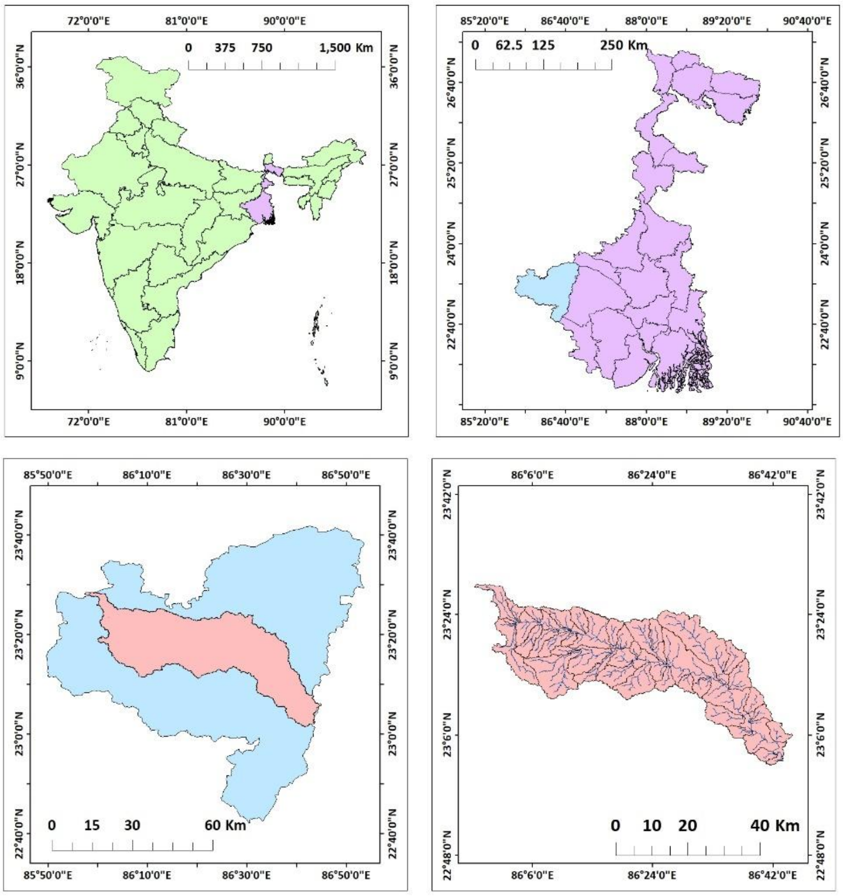

Figure 1: Location map of Kangsabati River basin in Purulia district

The Kangsabati River basin is prone to soil erosion due to various factors, including steep slopes, intense rainfall, and human activities. The combination of these factors can lead to severe soil loss, which has significant implications for agricultural productivity, water quality, and ecological balance. Therefore, it is crucial to assess the soil loss potential of the basin and implement appropriate conservation measures to mitigate the impacts of erosion.

To address the issue of soil erosion, it is essential to adopt sustainable land management practices, such as contour farming, terracing, and agroforestry. Additionally, afforestation and reforestation can help to stabilize the soil and reduce runoff. By implementing these measures, we can protect the valuable soil resources of the Kangsabati River basin and ensure its long-term sustainability.

3. DATA SOURCES AND MEHODOLOGY:

DATA SOURCES:

The primary data source for this study is the Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) Digital Elevation Model (DEM) acquired from the U.S. Geological Survey (USGS). This high-resolution DEM, with a spatial resolution of 30 meters, provides detailed topographic information essential for morphometric analysis. The ASTER GDEM was chosen due to its global coverage, relatively high accuracy, and free availability.

METHODOLOGY:

The ASTER DEM was used for estimation of morphometric parameter of Mula river basin such as: Drainage density (*Dd*), Bifurcation ratio (*Rb*), Stream length ratio (*Rl*), Mean stream length (*Lμ*), Stream order (*μ*), Stream frequency (*Fs*), Elongation ratio (*Re*), Circulatory ratio (*Rc*), Form factor (*Ff*), Basin relief (*Rr*), Relief ratio (Rr), Ruggedness Number (Rn), Compound Parameter Value (CP) etc. The table explains the formulas used for quantitative determination of morphometric parameters. Prioritization rating of five sub-watershed of Kangsabati River is carried out through ranking the computed morphological parameters. All the analysis was performed in the Geographical Information System environment with the aid of ArcGIS software.

Sr.No.	Morphometric parameters	Formula/Definition	References
Estimation of linear parameters			
1	Stream order (μ)	Ranking hierarchically	Strahler (1964)
2	Stream length (L_μ)	Total Length of the stream segments of that particular order	Horton(1945)
3	Mean stream length ($\bar{L}$)	$\bar{L} = \Sigma L_\mu / N_\mu$	Strahler (1964)
4	Stream length ratio (R_l)	$R_l = \bar{L}/\bar{L}_{(\mu-1)}$ Where $\bar{L}$ = The mean length of all stream segments of a given order(μ) $\bar{L}_{(\mu-1)}$ = The mean length of all stream segments of one order less to given order(μ)	Horton(1945)
5	Bifurcation ratio(R_b)	$R_b = N_\mu / N_{(\mu+1)}$ Where $N\mu$ = Total Number of stream segments of the order "μ" $N_{(\mu+1)}$ = Number of stream segments of the next higher order	Schumm (1956)
6	Drainage density (D_d)	$D_d = \Sigma L_\mu / A$ Where, ΣL_μ = Total Length of the stream segments of all Orders A = Area of the river basin or grid(km²)	Horton (1932)
Estimation of areal parameters			
1	Circularity ratio (R_c)	$R_c = 4\pi A/P^2$ Where A = Area of the basin (km²), P = Perimeter (km)	Miller (1953)Strahler (1964)
2	Elongation ratio (R_e)	$R_e = D/L = \frac{1.128\sqrt{A}}{L}$ Where D = diameter of a circle of the same area (A) as the basin, A = Area of the basin (km²), L = Basin length (km)	Schumm(1956)
3	Form Factor (F_f)	$F_f = A/L^2$ Where A = Area of the basin (km²), L = Basin length (km)	Horton(1932, 1945)
Estimation of relief aspects			
1	Basin relief (R)	$R = H - h$ Where H = maximum elevation (m), h = minimum elevation (m)	Hadley and Schumm (1961)
2	Relief ratio (R_r)	$R_r = R/L$ Where R = Basin relief, L = longest axis of major river (Basin length) in km	Schumm(1956)

By combining the results of morphometric analysis and soil loss modeling, we can identify areas with high soil loss potential and prioritize them for conservation measures. This information can be used to develop targeted strategies for soil erosion control and sustainable land management.

TABLE FOR SUB/MICRO WATERSHED PRIORITIZATION

WS_ID	Drainage Density	Bifurcation Ratio	Stream_Length	Stream_Frequency	Stream Oder	Elongated Ratio(Re)	Circularity Ratio(Rc)	Form Factor(Rf)	Basin Relief(H)	Relief Ratio(Rr)	Ruggedness Number(Rn)	Compound Parameter Value(CP)	Rank
1	82701.03	27567.12	0.00	0.00	0	26.33	0.07	0.01	-102.00	-32442.24	0.00	7775.03	W-2
2	82680.37	27560.24	0.00	0.00	0	26.33	0.07	0.01	-96.00	-30533.90	0.00	7963.71	W-4
3	13.21	12.72	2.67	0.05	1	30.58	422.92	33.67	50.00	233.44	13693295.76	1369409.50	W-25
4	19.01	14.14	11.68	0.02	4	27.68	293.91	23.40	60.00	304.16	14397532.84	1439828.68	W-26
5	82629.21	27543.18	3.93	0.02	1	26.34	0.07	0.01	-90.00	-28625.29	0.00	8148.75	W-5
6	41900.08	13966.86	0.09	0.02	1	24.59	0.13	0.01	40.00	8460.03	8858893.75	892328.56	W19
7	907.92	303.61	1.80	0.02	0	29.97	6.15	0.49	59.00	2238.65	16359362.37	1636291.00	W-28
8	82601.15	27533.83	1.30	0.02	0	26.34	0.07	0.01	-55.00	-17493.39	0.00	9261.43	W-7
9	17676.37	5892.37	0.09	0.02	0	26.21	0.32	0.03	40.00	5856.83	8460459.22	848995.14	W-18
10	82595.50	27531.95	2.67	0.02	0	26.34	0.07	0.01	-62.00	-19719.82	0.00	9037.47	W-6
11	82578.52	27526.29	1.54	0.02	0	26.35	0.07	0.01	-96.00	-30533.75	0.00	7950.30	W-3
12	23.30	14.94	11.27	0.02	4	29.15	239.78	19.09	24.00	141.88	5251719.96	525222.34	W-13
13	7.00	16.87	5.38	0.03	7	27.26	797.67	63.51	67.00	203.07	13241194.53	1324238.23	W-23
14	16.60	14.38	1.30	0.04	11	27.44	336.44	26.79	27.00	126.81	5363581.72	536415.85	W-14
15	9.53	20.55	2.84	0.04	3	20.34	585.98	46.65	88.00	232.21	17081538.52	1708254.47	W-29
16	258.53	88.99	0.00	0.00	0	22.36	21.61	1.72	32.00	483.27	4037653.51	403856.20	W-11
17	17.01	16.33	0.00	0.00	0	25.02	328.39	26.15	42.00	182.09	6081062.27	608169.93	W-15
18	20.90	16.56	0.00	0.00	0	22.36	267.27	21.28	54.00	231.92	10092701.52	1009333.58	W-21
19	5.29	20.50	11.27	0.01	0	24.69	1056.04	84.08	82.00	195.60	15364718.81	1536619.83	W-27
20	13.27	15.75	3.09	0.01	4	24.47	421.07	33.52	59.00	220.90	9306964.17	930775.52	W-20
21	2.50	25.37	6.99	0.01	11	28.65	2234.77	177.93	38.00	72.33	6544201.71	654678.83	W-17
22	41.33	21.24	1.06	0.01	5	21.92	135.15	10.76	86.00	509.07	18042016.23	1804284.28	W-31
23	82514.75	27505.03	0.13	0.01	1	26.36	0.07	0.01	-26.00	-8269.46	0.00	10175.09	W-8
24	11.58	15.43	0.93	0.01	1	26.22	482.50	38.42	32.00	119.92	4605459.05	460618.60	W-12
25	7.75	21.21	11.71	0.01	7	24.24	720.55	57.37	40.00	113.42	6300977.42	630197.37	W-16
26	27366.13	9122.28	1.64	0.01	0	22.76	0.20	0.02	-2.00	-316.33	-254558.44	-21836.37	W-1
27	82496.57	27498.97	0.00	0.00	5	26.36	0.07	0.01	-16.00	-5088.92	0.00	10491.71	W-9
28	28.23	18.24	0.09	0.05	3	23.50	197.87	15.75	66.00	346.27	10331185.55	1033188.15	W-22
29	82466.20	27488.85	1.31	0.05	0	26.36	0.07	0.01	0.00	0.00	0.00	10998.29	W-10
30	11.67	19.10	0.18	0.05	8	22.91	478.61	38.11	97.00	318.91	18831062.63	1883204.92	W-32
31	3.85	27.15	9.64	0.02	20	24.75	1451.93	115.60	100.00	203.94	17805239.08	1780717.60	W-30
32	16.45	14.62	0.09	0.02	3	27.87	339.54	27.03	109.00	517.67	19065972.44	1906702.47	W-33
33	2.40	37.69	0.96	0.02	28	21.67	2329.50	185.47	93.00	131.11	13330561.85	1333336.37	W-24

The study effectively utilized the ASTER DEM as the primary data source to conduct a comprehensive morphometric analysis of the Kangsabati River Basin. The high-resolution DEM provided accurate topographic information, enabling the precise delineation of watersheds and the calculation of key morphometric parameters. By combining GIS techniques and remote sensing data, this study offers a robust approach to assess soil loss potential and identify areas vulnerable to erosion. The integration of morphometric analysis with other relevant factors will provide a more comprehensive understanding of soil erosion processes and facilitate the development of targeted conservation strategies.

4. RESULTS AND DISCUSSION:

DEM:

The Digital Elevation Model (DEM) of the Kangsabati River Basin provides valuable insights into the topographic characteristics of the region. The DEM clearly depicts the undulating nature of the terrain, with a range of elevations from approximately 131 meters to 651 meters. The higher elevations are concentrated in the western and northern parts of the basin, while the lower elevations are found in the eastern and southern parts. The basin exhibits a dendritic

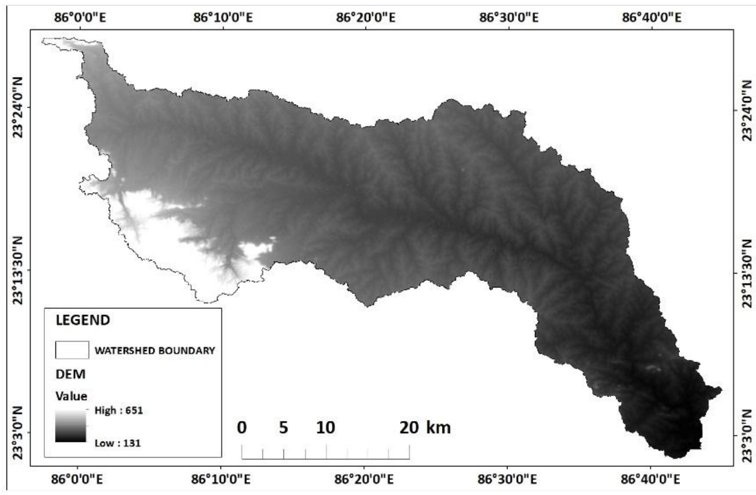

Figure 2: DEM of KangsabatiRiver basin in Purulia district

drainage pattern, with numerous tributaries feeding the main Kangsabati River. The steep slopes in the upper reaches of the basin, coupled with the high rainfall intensity, contribute to higher erosion potential in these areas. The lower reaches of the basin, with gentler slopes and lower elevations, are generally less prone to erosion.

Map for prioritization of the sub/micro watersheds using morphometric analysis:

The provided map offers a detailed view of the Kangsabati River basin, delineating its sub-watersheds and highlighting the intricate network of streams. The dendritic drainage pattern, characterized by branching tributaries, is evident, influencing the hydrological behaviour of the basin.

The sub-watersheds, demarcated by drainage divides, exhibit a wide range of sizes and shapes. Smaller, steeper sub-watersheds, often located in the upper reaches of the basin, are particularly susceptible to soil erosion due to the concentrated flow of water and the erosive power of the runoff. Conversely, larger, flatter sub-watersheds in the lower reaches may be less prone to erosion but are susceptible to flooding.

Morphometric analysis, a quantitative technique used to assess the form and spatial pattern of landforms, provides valuable insights into the erosion potential of these sub-watersheds. Parameters such as drainage density, stream frequency, and bifurcation ratio are crucial in understanding the hydrological processes and identifying areas prone to soil erosion.

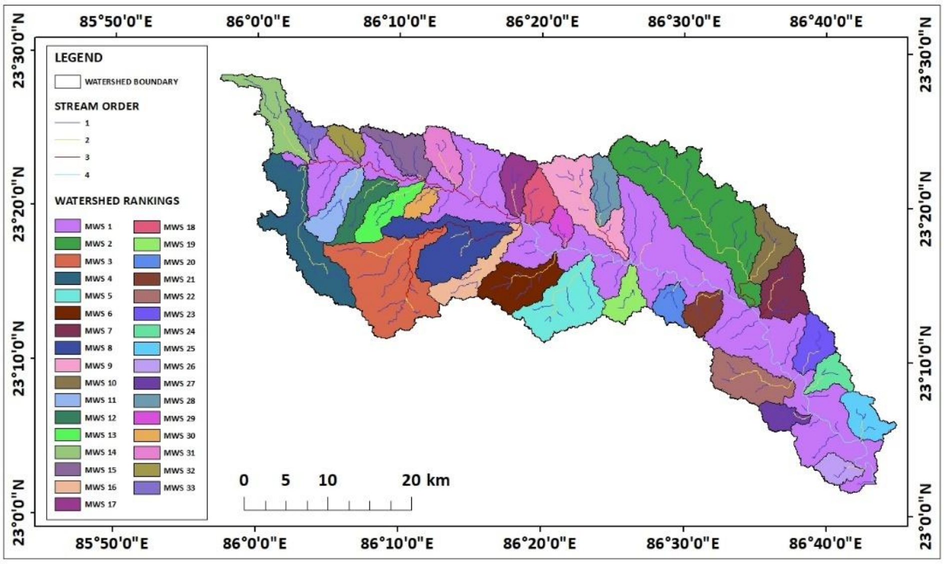

Figure 3: Map for prioritization of the sub/micro watersheds using morphometric analysis

Higher drainage density indicates a denser network of streams, leading to increased runoff and erosion. Stream frequency, the number of streams per unit area, also influences erosion potential. A higher stream frequency can result in greater surface runoff and soil erosion. Bifurcation ratio, the ratio of the number of streams of a given order to the number of streams of the next higher order, provides information about the branching pattern of the stream network. A higher bifurcation ratio suggests a more dendritic drainage pattern, which can be associated with higher erosion rates.

By integrating these morphometric parameters with other relevant factors like soil type, land use, and climate data, it is possible to develop a comprehensive assessment of soil loss potential. This information can be used to prioritize sub-watersheds for targeted conservation measures. For instance, sub-watersheds with high erosion potential may require afforestation, terracing, or other soil conservation practices to reduce runoff and prevent soil erosion.

In conclusion, the morphometric analysis of the Kangsabati River basin provides valuable insights into the hydrological processes and erosion susceptibility of the region. By understanding the spatial variability of these factors, it is possible to implement targeted conservation measures to mitigate soil erosion and protect the basin's valuable resources.

5. CONCLUSION:

The morphometric analysis of the Kangsabati River basin provides valuable insights into the hydrological processes and erosion susceptibility of the region. The delineated sub-watersheds, characterized by their unique topographic and hydrologic features, exhibit varying degrees of erosion potential. The dendritic drainage pattern, characterized by a branching network of streams, plays a significant role in shaping the hydrological response of the basin. Higher-order streams, with their greater erosive power, can transport significant amounts of sediment, contributing to downstream deposition and channel aggradation.

The morphometric parameters, such as drainage density, stream frequency, and bifurcation ratio, provide quantitative measures of the basin's hydrological characteristics. Higher drainage density, indicating a denser network of streams, is associated with increased runoff and erosion potential. Similarly, higher stream frequency and lower bifurcation ratios can contribute to higher erosion rates. By understanding the spatial variability of these parameters, it is possible to identify sub-watersheds that are more vulnerable to soil erosion. These vulnerable areas often exhibit steep slopes, high drainage densities, and low vegetation cover. Implementing targeted conservation measures, such as afforestation, terracing, and contour farming, can help to mitigate soil erosion and protect the basin's valuable resources.

Furthermore, integrating the morphometric analysis with other relevant factors, such as soil type, land use, and climate data, can provide a more comprehensive assessment of soil erosion risk. This integrated approach can help to identify areas that are most susceptible to erosion and prioritize conservation efforts accordingly. In conclusion, the morphometric analysis of the Kangsabati River basin provides a valuable tool for understanding the hydrological processes and erosion susceptibility of the region. By identifying vulnerable areas and implementing appropriate conservation measures, it is possible to protect the basin's valuable resources and ensure the long-term sustainability of the ecosystem.

THEMATIC REPRESENTATION OF THE SPATIAL DISTRIBITION OF WATER QUALITY INDEX: A CASE STUDY ON PURULIA, WEST BENGAL

1. INTRODUCTION:

Quality of drinking water indicates water acceptability for human consumption. Water quality depends on water composition influenced by natural process and human activities. Water quality is characterized on the basis of water parameters (physical, chemical, and microbiological), and human health is at risk if values exceed acceptable limits. Various agencies such as the World Health Organization (WHO) and Centers for Disease Control (CDC) set exposure standards or safe limits of chemical contaminants in drinking water. A common perception about water is that clean water is good-quality water indicating.

Water Quality Index (WQI) is considered as the most effective method of measuring water quality. A number of water quality parameters are included in a mathematical equation to rate water quality, determining the suitability of water for drinking. The index was first developed by Horton in 1965 to measure water quality by using 10 most regularly used water parameters. The method was subsequently modified by different experts. These indices used water quality parameters which vary by number and types. The weights in each parameter are based on its respective standards, and the assigned weight indicates the parameter's significance and impacts on the index. A usual WQI method follows three steps which include

(1) selection of parameters,

(2) determination of quality function for each parameter,

(3) aggregation through mathematical equation.

The index provides a single number that represents overall water quality at a certain location and time based on some water parameters. The index enables comparison between different sampling sites. WQI simplifies a complex dataset into easily understandable and usable information. The water quality classification system used in the WQI denotes how suitable water is for drinking. The single-value output of this index, derived from several parameters, provides important information about water quality that is easily interpretable, even by lay people.

2. STUDY AREA:

Purulia district, located in the westernmost region of West Bengal, India, lies between 22°42'35"N to 23°42'0"N latitude and 85°49'25"E to 86°54'37"E longitude. This district spans an area of approximately 6,259 square kilometers and is characterized by an undulating topography with plateaus, hills, and interspersed plains. Purulia's landscape forms a part of the Chotanagpur Plateau, making it geologically significant due to its abundance of hard rocks and red lateritic soils.

The district is bordered by Bankura and West Midnapore to the east, Jharkhand state to the west and north, and the state of Odisha to the south. The Kangsabati, Subarnarekha, and Damodar rivers are the primary water sources in this region, supporting both agricultural and domestic needs. However, the water quality is increasingly under scrutiny due to anthropogenic influences such as industrial discharges, agricultural runoff, and improper waste disposal.

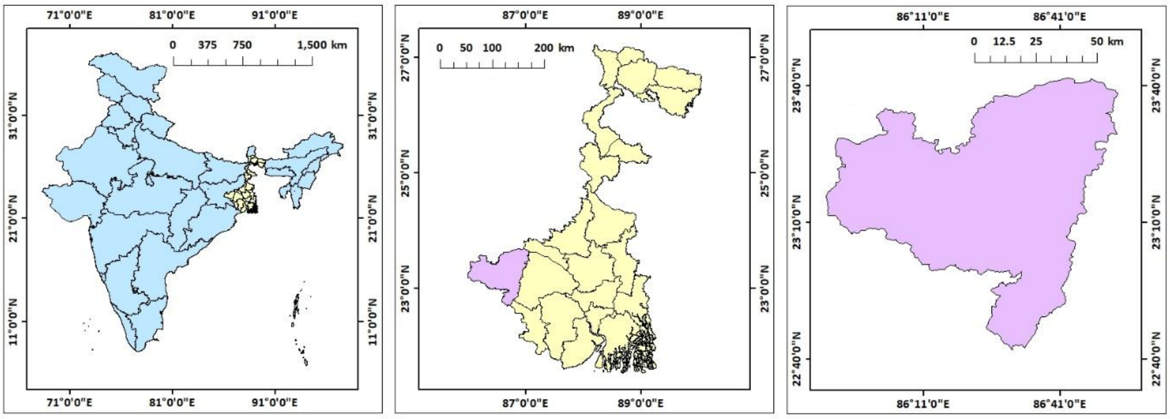

Figure 4: Location map of Purulia district

Purulia experiences a tropical climate, with hot summers, monsoonal rains, and cool winters. The annual rainfall averages around 1100–1500 mm, primarily during the monsoon season (June to September). Despite sufficient rainfall, water scarcity during non-monsoon months is prevalent due to the poor water retention capacity of the soil and uneven distribution of resources.

The socio-economic fabric of Purulia is predominantly rural, with agriculture being the main occupation. However, industrial activities, particularly in mineral-based sectors, are rising in the region. This shift has created a dual challenge of managing water resources and addressing pollution issues.

Given its ecological and socio-economic importance, understanding the spatial distribution of water quality is crucial for sustainable development and resource management in Purulia. This study uses thematic mapping to evaluate water quality status and its spatial variation across different locations within the district.

3. DATA SOURCES AND METHODOLOGY:

Weighted arithmetic Water Quality Index (WQI) method:

The weighted arithmetic WQI method was applied to assess water suitability for drinking purposes. In this method, water quality rating scale, relative weight, and overall WQI were calculated by the following formula: **$q_i = (C_i / S_i) \times 100$**

where q_i, C_i, and S_i. indicated quality rating scale, concentration of i parameter, and standard value of / parameter, respectively.

Relative weight was calculated by: **wi = 1 / Si**

where the standard value of the i parameter is inversely proportional to the relative weight.

Finally, overall WQI was calculated this formula: **WQI = ∑Zqiwi / ∑Ewi**

4. RESULTS AND DISCUSSION:

The provided tables summarize the water quality index (WQI) data for Purulia, categorizing locations based on five water quality classes: "Very Safe," "Safe," "Moderate," "Unsafe," and "Very Unsafe." The tabulated data indicates a clear prevalence of poor water quality across the district, with a significant number of locations falling into the "Unsafe" and "Very Unsafe" categories. Conversely, fewer locations are categorized as "Very Safe" or "Safe," emphasizing the uneven distribution of water quality.

The numerical data aligns with the thematic maps, reinforcing the dominance of suboptimal water quality in specific regions. The central and western blocks display the highest concentration of poor-quality water, likely driven by localized anthropogenic activities and geogenic factors. The tables further highlight variability in water quality, underscoring the need for targeted interventions to improve the status of water resources in Purulia while addressing potential health hazards to the affected communities.

OBJECTID	BLOCK	VALUE_1	VALUE_2	VALUE_3	VALUE_4	VALUE_5	VERY UNS/	UNSAFE	MODERAT	SAFE	VERY SAFE
1	Arsha	2.49E+08	96210900	4158900	1172700	43795800	2.49E+08	96210900	43795800	4158900	1172700
2	Bagmundi	900	2.54E+08	0	1.69E+08	0	2.54E+08	1.69E+08	900	0	0
3	Balarampu	4716900	69088500	0	1.01E+08	1.49E+08	1.49E+08	1.01E+08	69088500	4716900	0
4	Barabazar	2.37E+08	51929100	5902200	1.08E+08	57816000	2.37E+08	1.08E+08	57816000	51929100	5902200
5	Bundwan	0	0	4E+08	0	0	4E+08	0	0	0	0
6	Hura	1.2E+08	1.31E+08	3774600	15198300	1.31E+08	1.31E+08	1.31E+08	1.2E+08	15198300	3774600
7	Jhalda 1	1.28E+08	60059700	0	1.84E+08	0	1.84E+08	1.28E+08	60059700	0	0
8	Jhalda 2	86950800	1.09E+08	61199100	20560500	15268500	1.09E+08	86950800	61199100	20560500	15268500
9	Joypur	74232900	0	0	1.04E+08	47975400	1.04E+08	74232900	47975400	0	0
10	Kasipur	1.96E+08	96565500	1.2E+08	0	27274500	1.96E+08	1.2E+08	96565500	1.2E+08	0
11	Manbazar	19978200	51440400	68712300	1.56E+08	76590900	1.56E+08	76590900	68712300	51440400	19978200
12	Manbazar	11656800	1.12E+08	1.48E+08	11098800	0	1.48E+08	1.12E+08	11656800	11098800	0
13	Neturia	0	77925600	81624600	229500	0	81624600	77925600	229500	0	0
14	Para	20304000	0	73854900	1.37E+08	95852700	1.37E+08	95852700	73854900	20304000	0
15	Puncha	57215700	13030200	46250100	38771100	1.75E+08	1.75E+08	57215700	46250100	38771100	13030200
16	Purulia 1	56621700	1.12E+08	3825900	42957900	68500800	1.12E+08	68500800	56621700	42957900	3825900
17	Purulia 2	16106400	27831600	1.27E+08	53727300	1.05E+08	1.27E+08	1.05E+08	53727300	27831600	16106400
18	Raghunath	2747700	22703400	91107000	36900	75439800	91107000	75439800	2747700	22703400	36900
19	Raghunath	0	17294400	0	51235200	1.18E+08	1.18E+08	51235200	17294400	0	0
20	Santuri	0	0	1.76E+08	0	0	1.76E+08	0	0	0	0

The provided maps and data reveal the spatial distribution of water quality in Purulia district, offering a comprehensive understanding of its current status. The thematic representation using pie charts highlights the proportional distribution of water quality classes across various locations. These categories—ranging from "Very Safe" to "Very Unsafe"—indicate significant variability in water quality. Notably, "Unsafe" and "Very Unsafe" categories dominate in many areas, underscoring the pressing concern of deteriorating water conditions in the district. While some pockets show a relatively higher proportion of "Safe" and "Moderate" water quality, these are sporadic and insufficient to offset the larger trend of unsafe water distribution.

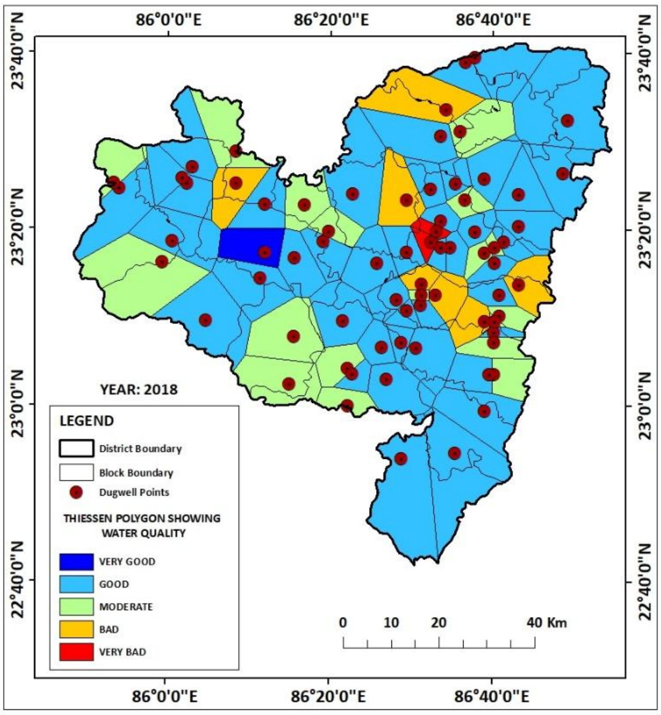

Figure 5: Water Quality mapping on Purulia district using Thiessen polygon

The Thiessen polygon map further refines this analysis by interpolating water quality indices across the district using dugwell points as reference locations. This spatial interpolation provides a detailed visualization of water quality zones, classified into five categories: "Very Good," "Good," "Moderate," "Bad," and "Very Bad." The majority of the district is characterized by "Moderate" water quality, followed by significant areas classified as "Bad." Only a few scattered locations exhibit "Very Good" water quality, reflecting localized resilience or better management practices in those areas. The central and eastern regions appear to fare slightly better compared to the western and southern parts, which are more adversely impacted.

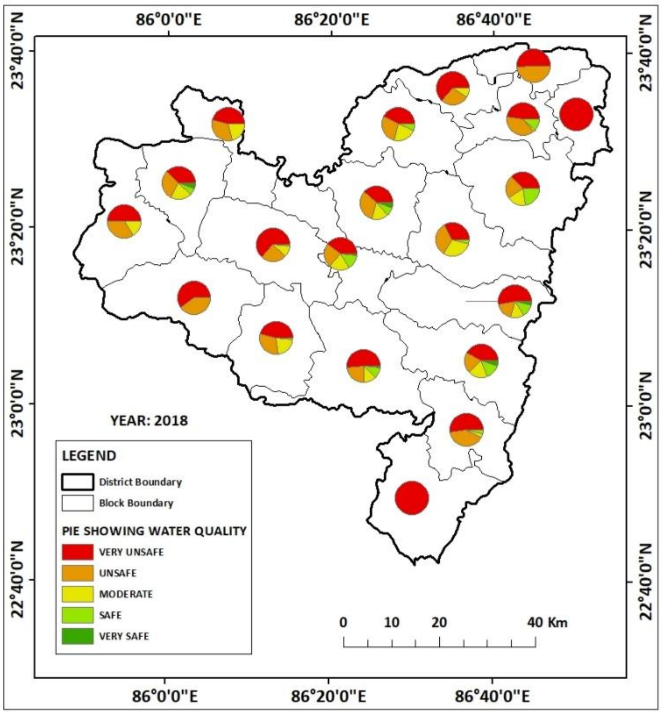

Figure 6: Water Quality mapping on Purulia district using Pie diagrams

The combined analysis of these maps suggests a stark disparity in water quality across Purulia. This variability can be attributed to several factors, including geological formations, land-use patterns, and anthropogenic activities. Areas with higher concentrations of "Unsafe" and "Bad" water quality are likely affected by industrial discharge, agricultural runoff, or inadequate sanitation facilities. Conversely, zones with "Very Good" or "Safe" water quality could be benefiting from better groundwater recharge conditions or minimal human-induced pollution.

It is evident from these findings that the water quality in Purulia is under stress, with a significant portion of the population potentially exposed to health risks. The predominance of "Unsafe" and "Bad" water quality calls for urgent intervention to mitigate the adverse impacts. Strategies such as community-level water treatment facilities, stricter regulation of industrial and agricultural pollutants, and awareness campaigns on sustainable water use are essential. Additionally, further studies integrating seasonal variations and groundwater recharge dynamics could provide a deeper understanding of the factors influencing water quality and inform long-term management strategies.

5. CONSLUSION:

The spatial analysis of water quality in Purulia district reveals a concerning pattern of uneven and largely suboptimal water quality across the region. The thematic maps and tabulated data illustrate that a significant portion of the district's water resources fall into "Unsafe" and "Very Unsafe" categories, indicating serious contamination issues and potential risks to public health. The limited areas classified as "Safe" or "Very Safe" highlight isolated instances of better water quality, which are insufficient to counterbalance the district-wide challenges.

The Thiessen polygon map further refines the spatial understanding of water quality by showing zones of influence for each monitoring point, revealing the widespread prevalence of "Moderate," "Bad," and "Very Bad" water quality. The data suggests that water pollution in Purulia is influenced by a combination of anthropogenic factors, such as industrial runoff, agricultural practices, and insufficient sanitation infrastructure, as well as natural conditions like geology and soil composition.

Given these findings, immediate measures are needed to improve water quality and mitigate health risks. Recommendations include implementing water treatment systems, improving waste management, regulating industrial and agricultural discharge, and promoting public awareness of sustainable water use. A long-term approach involving continuous monitoring, policy enforcement, and integrated water resource management is essential to ensure sustainable development in Purulia.

THEMATIC REPRESENTATION OF RUNOFF DEPTH USING SCS CN METHOD: A CASE STUDY ON RANIBAND, BANKURA, WEST BENGAL

1. INTRODUCTION:

The assessment of surface runoff is a critical component in hydrological studies, especially in regions where land use and land cover (LULC) dynamics significantly influence water availability and management. Among various methodologies, the Soil Conservation Service Curve Number (SCS-CN) method stands out as a widely adopted empirical approach for estimating runoff depth. This method integrates key hydrological factors such as soil type, land use, hydrological condition, and antecedent moisture to predict the volume of runoff generated by a given rainfall event. Its simplicity, effectiveness, and adaptability make it particularly valuable for analysing diverse LULC scenarios.

Land use and land cover play a pivotal role in determining the rate and volume of surface runoff. Urbanized areas with impervious surfaces, agricultural lands, forests, and barren lands exhibit distinct runoff characteristics. By assigning specific Curve Numbers (CN) to these elements, the SCS-CN method quantifies the runoff depth, enabling planners to evaluate how different land cover types respond to precipitation events. This information is essential for understanding flood risks, designing water conservation structures, and implementing sustainable watershed management practices.

The integration of geospatial technologies, such as Geographic Information Systems (GIS), further enhances the efficiency and precision of runoff estimation. GIS tools facilitate the spatial analysis and thematic representation of runoff depth by mapping the variation in runoff across a given watershed. The ability to visually represent this data aids in identifying critical areas prone to waterlogging, erosion, or inadequate drainage. Such thematic maps serve as a foundation for decision-making in agricultural planning, urban development, and environmental conservation.

By calculating the runoff depth for various LULC elements, this assignment seeks to provide insights into the hydrological behaviour of a specific region. It emphasizes the interplay between rainfall, land characteristics, and surface runoff, offering a comprehensive understanding of water movement within the landscape. The outcomes will contribute to developing effective strategies for managing water resources, mitigating flood risks, and promoting land sustainability.

In this assignment, the practical application of the SCS-CN method combined with GIS-based thematic representation not only demonstrates the importance of hydrological modelling but also highlights its relevance to real-world challenges. The results will underline the need for an integrated approach to water resource management, aligning hydrological analysis with spatial planning to ensure a balanced and sustainable utilization of natural resources.

2. STUDY AREA:

The study area selected for this assignment is the Ranibandh block, located in the Bankura district of West Bengal, India. Geographically, Ranibandh lies between approximately **22°58'N to 23°12'N latitude** and **86°45'E to 86°59'E longitude**, covering a region characterized by a mix of forested landscapes, agricultural lands, and scattered settlements. It is situated in the western part of Bankura, bordering the Purulia district and the state of Jharkhand, making it a transitional zone between the Chota Nagpur Plateau and the plains of Bengal.

Ranibandh is part of the semi-arid and drought-prone regions of West Bengal, with a tropical climate marked by distinct wet and dry seasons. The area experiences an average annual rainfall of approximately 1,200 mm, primarily during the monsoon months from June to September. However, erratic rainfall patterns and high evaporation rates often contribute to water scarcity and seasonal drought conditions, affecting agricultural productivity and water availability.

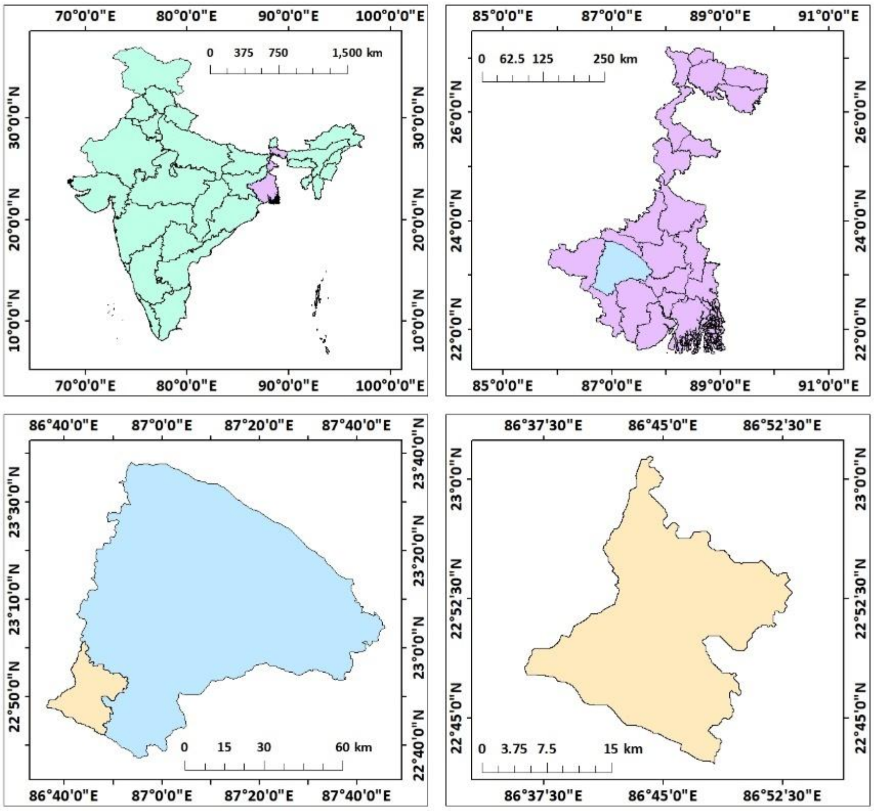

Figure 7: Location map of Ranibandh

The block is predominantly rural, with agriculture and forest-based livelihoods forming the economic backbone of the local population. Paddy cultivation is the primary agricultural activity, complemented by pulses, oilseeds, and vegetables. The region's forests, consisting of sal and other deciduous species, play a crucial role in supporting biodiversity and providing resources for local communities.

Ranibandh's topography is gently undulating, interspersed with hills, rocky outcrops, and rivers such as the Kansabati and Kumari, which are vital for irrigation and water supply. The soil types vary from sandy loam to lateritic, influencing land use patterns and hydrological responses. This diversity in land cover and topography makes Ranibandh an ideal site for applying the SCS-CN method to assess runoff dynamics. Understanding runoff patterns here is essential for effective water resource management and sustainable agricultural practices.

3. DATA SOURCES AND METHODOLOGY:

DATA SOURCES:

The data sources utilized for this study are diverse and comprehensive, ensuring the accuracy and relevance of the runoff depth estimation using the SCS-CN method. The key data sources include:

- **Soil Texture Data (Hydrological Soil Group or HSG)**:
    - Obtained from National Bureau of Soil Survey and Land Use Planning (NBSSLUP) for detailed soil classifications and hydrological properties.
    - Additional soil information sourced from the Food and Agriculture Organization (FAO) Soil Database, providing global context and compatibility.
    - The Bhukosh Portal, an initiative of the Geological Survey of India, offers spatial soil maps to validate and enhance the dataset.
- **Land Use Land Cover (LULC) Data**:
    - Acquired from the ESRI Sentinel-2 Land Cover Explorer, which utilizes Sentinel-2 satellite imagery to provide global land cover maps.
    - The dataset includes detailed classification of land cover types such as forests, agriculture, and urban areas, crucial for assigning Curve Numbers (CN).

These datasets collectively ensure a robust analysis of runoff depth by integrating accurate spatial and temporal data across soil, rainfall, and land cover parameters. Each source has been selected based on its reliability, resolution, and relevance to the study area, enabling a comprehensive evaluation of hydrological behaviour in the Ranibandh block.

METHODOLOGY:

- **SCS CN:** The Soil Conservation Service Curve Number (SCS- CN) method is widely used for predicting direct runoff volume for a given rainfall event. This method was originally developed by the US Department of Agriculture, Soil Conservation Service. The runoff estimated from SCS model primarily depends upon rainfall pattern and antecedent moisture condition. Especially for small and medium sized basins, CN curve number method gave good correlation between the measured and estimated runoff. In Indian scenario, as there is lack of runoff records at sub/micro watershed level, SCS curve number methods are widely used to compute run-off. SCS curve number method combines the watershed parameters and climatic factors in one entity called "the Curve Number" (CN). Many researchers have utilized the

Geographic Information System (GIS) technique for runoff estimation by Curve Number value 'CN'.

- **CURVE NUMBER:** Curve Number is a function of antecedent moisture content and slope. Antecedent Moisture Condition (AMC) is an indicator of watershed wetness and availability of soil moisture storage prior to a storm, and can have a significant effect on runoff volume. Three levels of AMC are used in the CN method: AMC-I for dry, AMC-II for normal, and AMC-III for wet conditions. The CNs as per AMCs are CN-I, CN-II and CN-III.
- **CN(I) = 4. 2 CN(II) / 10 - 0.058CN(II)**
- **CN(III) = 23 CN(II) / 10 + 0.13CN(II)**
- **CN = ∑ (CNi × Ai) / A**

  (CN = weighted Curve Number, Cni = Curve Numbers from 1 to any number of N, Ai= Area with curve number CNi, A= Area of the unit under study.)

LIST OF CN-II

Sr. No.	Landuse	Treatment/ Practice	Hydrologic Condition	Hydrologic Soil Group			
				A	B	C	D
(1)	Cultivated	Straight Row	Poor	72	81	88	91
	(Row Crops)	Straight Row	Good	67	78	85	89
		Contoured	Poor	70	79	81	86
		Contoured	Good	65	75	82	86
		Contoured & Terraced	Poor	66	74	80	82
		Contoured & Terraced	Good	62	71	78	81
		Paddy	-	95	95	05	95
(2)	Forest	Dense	-	26	40	58	61
		Open	-	28	44	60	64
		Shrubs	-	33	47	64	67
(3)	Pasture	-	Poor	68	79	86	89
		-	Fair	48	69	79	84
		-	Good	39	61	71	80
(4)	Waste Land	-	-	71	80	85	88
(5)	Water bodies	-	-	100	100	100	100
(6)	Settlement	-	-	61	75	83	87

- **AMC (Antecedent Moisture Condition):**
  Antecedent soil moisture is known to have significant effect on both the volume rate of runoff. Recognizing that it is a significant factor, SCS developed three antecedent soil moisture conditions, which are labelled as I (dry), II (Average), III (Wet) soil conditions for each are follows –

AMC group	Specification	Rainfall in dormant season (mm)	Rainfall in growing season (mm)
AMC-1 (Dry)	Soils are dry but not to wilting point, satisfactory cultivation has taken place.	<12.5	<35.6
AMC-II (Average)	This level represents the average conditions	12.5 – 27.5	35.6 – 53.3
AMC-III (Wet)	Heavy rainfall or light rainfall and low temperature have occurred within last 5 days, which situates the soils.	<27.5	>53.3

CALCULATION TABLE FOR RUNOFF:

ID	GRIDCODE	Class_Name	FAOSOIL	TEXTURE	HSG	CN	S	Q
2	6	agricultural	Lf32-1b	SANDY_LOAM	A	72	98.77777778	7.039164248
2	6	agricultural	Lf96-2ab	SANDY_CLAY_LOAM	C	88	34.63636364	18.75586224
2	6	agricultural	I-Ne	LOAM	B	81	59.58024691	12.37800231
2	6	agricultural	I-Ne	LOAM	B	81	59.58024691	12.37800231
3	10	dense forest	Lf32-1b	SANDY_LOAM	A	26	722.9230769	1.5099141
3	10	dense forest	Lf96-2ab	SANDY_CLAY_LOAM	C	58	183.9310345	2.271395729
3	10	dense forest	I-Ne	LOAM	B	40	381	0.009428049
3	10	dense forest	I-Ne	LOAM	B	40	381	0.009428049
4	7	dry fallow	Lf32-1b	SANDY_LOAM	A	71	103.7464789	6.580561137
4	7	dry fallow	Lf96-2ab	SANDY_CLAY_LOAM	C	85	44.82352941	15.70182698
4	7	dry fallow	I-Ne	LOAM	B	80	63.5	11.65540401
4	7	dry fallow	I-Ne	LOAM	B	80	63.5	11.65540401
6	3	hsp	Lf32-1b	SANDY_LOAM	A	61	162.3934426	3.032803285
6	3	hsp	Lf96-2ab	SANDY_CLAY_LOAM	C	83	52.02409639	13.94665711
6	3	hsp	I-Ne	LOAM	B	75	84.66666667	8.557238478
6	3	hsp	I-Ne	LOAM	B	75	84.66666667	8.557238478
1	5	agri fallow	Lf32-1b	SANDY_LOAM	A	71	103.7464789	6.580561137
1	5	agri fallow	Lf96-2ab	SANDY_CLAY_LOAM	C	85	44.82352941	15.70182698
1	5	agri fallow	I-Ne	LOAM	B	80	63.5	11.65540401
1	5	agri fallow	I-Ne	LOAM	B	80	63.5	11.65540401
5	8	exposer	Lf32-1b	SANDY_LOAM	A	100	0	40
5	8	exposer	Lf96-2ab	SANDY_CLAY_LOAM	C	100	0	40
5	8	exposer	I-Ne	LOAM	B	100	0	40
5	8	exposer	I-Ne	LOAM	B	100	0	40
7	4	open forest	Lf32-1b	SANDY_LOAM	A	28	653.1428571	1.020681585
7	4	open forest	Lf96-2ab	SANDY_CLAY_LOAM	C	60	169.3333333	2.765442365
7	4	open forest	I-Ne	LOAM	B	44	323.2727273	0.177886546
7	4	open forest	I-Ne	LOAM	B	44	323.2727273	0.177886546
8	9	water	Lf32-1b	SANDY_LOAM	A	100	0	40
8	9	water	Lf96-2ab	SANDY_CLAY_LOAM	C	100	0	40
8	9	water	I-Ne	LOAM	B	100	0	40
8	9	water	I-Ne	LOAM	B	100	0	40

4. RESULTS AND DISCUSSION:

The results section presents the spatial distribution of land use and land cover (LULC) and soil texture within the Ranibandh subdivision. The discussion section delves into the interpretation of these maps, highlighting their implications for runoff generation and soil erosion. It analyses the relationship between LULC, soil texture, and runoff potential, and discusses the potential impacts on water resources and agricultural productivity.

LULC map:

The map depicts the spatial distribution of Land Use and Land Cover (LULC) classes within the Ranibandh subdivision. LULC plays a crucial role in determining runoff generation as different land cover types have varying infiltration capacities. The basin exhibits a diverse range of LULC classes, including agricultural land, agricultural fallow land, dense forest, dry fallow land, exposed areas, built-up areas, open forest, and water bodies. Permeable surfaces like dense forests and agricultural lands generally have higher infiltration rates, resulting in lower runoff. Conversely, impervious surfaces like built-up areas and exposed lands have lower infiltration rates, leading to higher runoff. Understanding the spatial distribution of LULC classes is essential for assessing runoff potential and implementing appropriate soil and water conservation measures. By identifying areas with high runoff potential, targeted interventions like afforestation, contour bunding, and terrace farming can be implemented to reduce runoff and soil erosion.

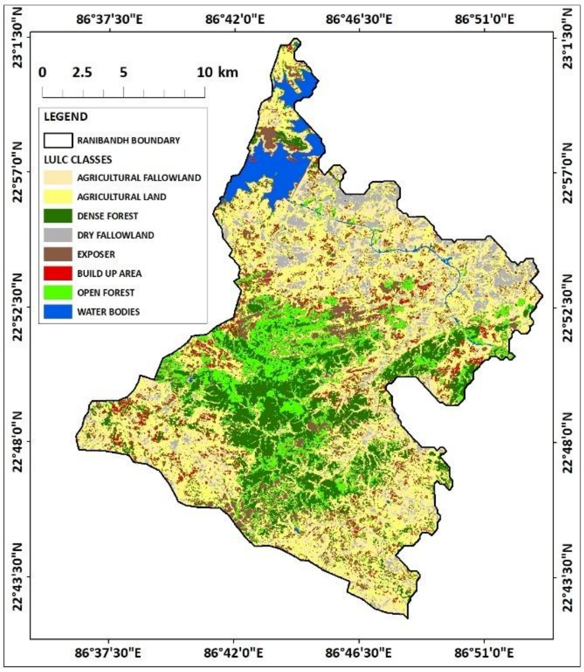

Figure 8: LULC map

Soil texture map:

The map depicts the spatial distribution of soil texture within the Ranibandh subdivision. Soil texture, characterized by the relative proportions of sand, silt, and clay, significantly influences soil infiltration capacity and runoff generation. The basin

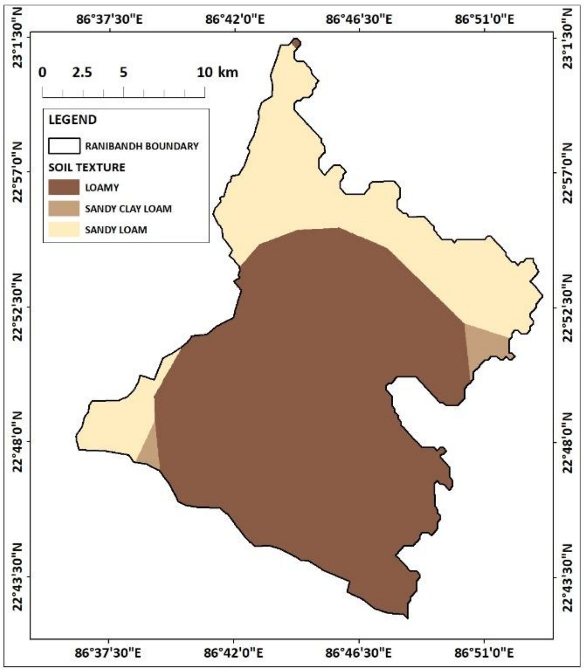

Figure 9: Soil texture map

exhibits a diverse range of soil textures, including loamy, sandy clay loam, and sandy loam soils. Loamy soils, with a balanced mixture of sand, silt, and clay, have moderate infiltration capacity and moderate runoff potential. Sandy clay loam soils, with a higher proportion of clay, generally have lower infiltration capacity and higher runoff potential. Sandy loam soils, with a higher proportion of sand, have higher infiltration capacity and lower runoff potential. Understanding the spatial distribution of soil texture is crucial for assessing runoff potential and implementing appropriate soil and water conservation measures. By identifying areas with low infiltration capacity, targeted interventions like contour bunding, terrace farming, and mulching can be implemented to reduce runoff and soil erosion.

Runoff map:

The spatial distribution of runoff depth across the Ranibandh block, as depicted in the runoff map, reveals significant variations that correlate closely with the underlying land use and land cover (LULC) patterns. The interplay between rainfall, soil type, and LULC elements influences the hydrological response of the region, resulting in distinct runoff behaviors in different zones.

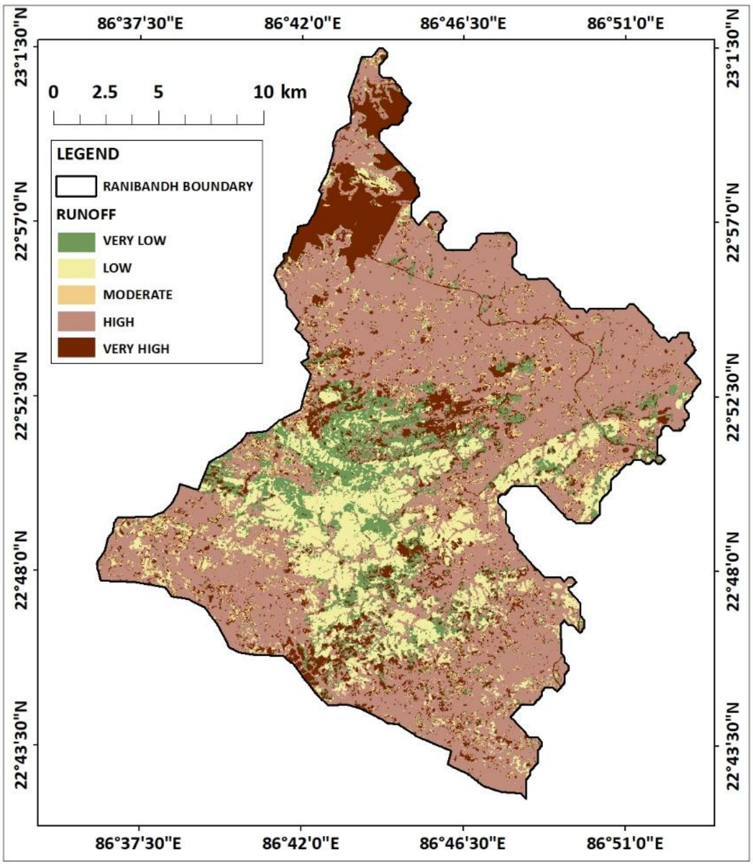

Figure 10: Runoff map of Ranibandh

Low runoff zones are associated with open forest areas and agricultural lands where vegetation cover, though not as dense as forests, still contributes to moderate infiltration. These zones are interspersed across the block, particularly surrounding dense forest regions. The agricultural practices in these areas, coupled with relatively permeable soils, help moderate runoff generation. However, the transition to more intensive land use in some parts of these zones could potentially alter the hydrological balance, highlighting the need for sustainable agricultural practices.

Moderate runoff zones are characterized by a mix of agricultural lands, fallow lands, and scattered barren areas. These regions, evident from the LULC map, reflect intermediate infiltration rates due to their heterogeneous land cover. While agricultural practices dominate, the absence of vegetation in fallow and barren lands reduces water retention, leading to moderate runoff. This indicates the potential for soil erosion and nutrient loss, emphasizing the importance of soil conservation measures such as cover cropping and contour bunding in these areas.

High runoff zones correspond to barren lands and buildup areas, as indicated by the LULC map. These regions, found predominantly in the northern and eastern parts of the block, experience significant surface runoff due to impervious surfaces and poor infiltration. The combination of rocky outcrops, settlements, and degraded lands exacerbates water flow, leading to increased runoff volumes. These zones are particularly susceptible to erosion and flooding, requiring interventions like afforestation, check dams, and improved drainage systems to mitigate runoff impacts.

Very high runoff zones are concentrated in specific areas where water bodies and highly degraded lands dominate. These regions, with minimal infiltration potential, experience rapid accumulation of runoff due to their impermeable nature. The presence of water bodies, while reducing surface flow locally, contributes to runoff aggregation in adjoining degraded areas. This highlights the importance of integrated watershed management approaches, such as enhancing vegetative cover in the buffer zones and controlling land degradation, to manage surface runoff effectively.

In summary, the runoff map, when correlated with the LULC map, provides a detailed understanding of the hydrological dynamics within Ranibandh block. The results emphasize the critical role of LULC in shaping runoff behavior, with dense forests and vegetation-rich areas acting as natural buffers, while barren and impervious surfaces contribute to heightened runoff. These findings underline the need for targeted land management strategies to mitigate hydrological risks and promote sustainable development in the region.

5. CONCLUSION:

The analysis of runoff patterns in the Ranibandh block, using the SCS-CN method and supported by LULC data, highlights the critical relationship between land use and hydrological behavior. Dense forests and vegetation-rich areas exhibit very low to low runoff due to their high infiltration capacities and natural water-retention properties. In contrast, barren lands, degraded areas, and impervious surfaces show high to very high runoff, leading to potential issues like soil erosion, flooding, and waterlogging.

The findings underline the importance of sustainable land use practices and ecosystem preservation in managing runoff. Forested regions, which act as natural runoff regulators, need to be conserved and expanded to maintain ecological balance and improve groundwater recharge. Agricultural zones require careful management, including soil conservation techniques, to minimize runoff and prevent degradation. High runoff areas, particularly in barren and built-up zones, demand interventions like afforestation, water harvesting structures, and improved drainage systems to mitigate hydrological risks.

This study demonstrates the value of integrating geospatial technologies with hydrological modelling to assess and manage surface runoff. By identifying critical areas requiring intervention, the results provide a foundation for implementing targeted strategies to enhance water resource management, mitigate environmental impacts, and ensure sustainable development in the Ranibandh block.